1

$m = 2000\ kg$ $\quad t = ?$

$v_0 = 80\ km/h = 22'22\ m/s$ $\quad F = ?$

$\Delta x = 75\ m$

a)

$$\Delta v = a\,t$$

$$-22'22 = a\,t$$

$$a = -\frac{22'22}{t}$$

$$\Delta x = v_0 t + \frac{1}{2} a t^2$$

$$75 = 22'22\,t + \frac{1}{2}\left(-\frac{22'22}{t}\right) t^2$$

$$75 = 22'22\,t - 11'11\,t$$

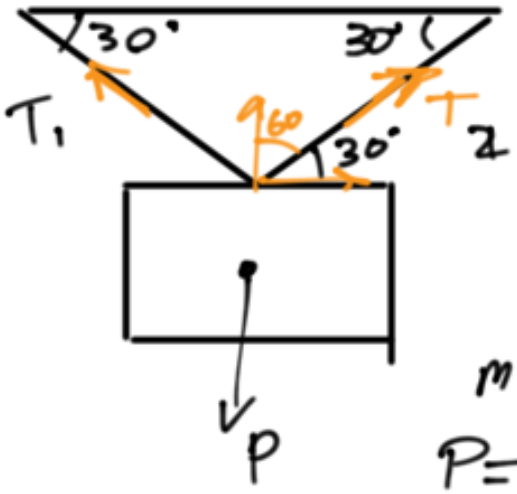
$$t = \frac{75}{22'22 - 11'11} = 6'75\ s$$

$$a = -\frac{22'22}{6'75} = -3'29\ m/s^2$$

$$F = m\,a = 2000 \cdot 3'29 = \underline{\underline{6584'7\ N}}$$

↳ En módulo

②



Max: 15 N

$g = 10\ m/s^2$

$m = 2kg$

$P = 20\ N$

$$\vec{P} = -mg\,\vec{j}$$

$$\vec{T_1} = T_x\,\vec{i} + T_y\,\vec{j} = -T_1 \cos\alpha\,\vec{i} + T_1 \sin\alpha\,\vec{j}$$

$$\vec{T_2} = T_2 \cos\alpha\,\vec{i} + T_2 \sin\alpha\,\vec{j}$$

$$\sum \vec{F} = m\,a$$

Eje x: $-T_1 \cos 30 + T_2 \cos 30 = 0$; $T_1 = T_2 = T$

Eje y: $T \sin\alpha + T \sin\alpha - mg = 0$

$$2T = 2 \cdot 10$$

$$T = 20N \text{ (Se rompe)}$$

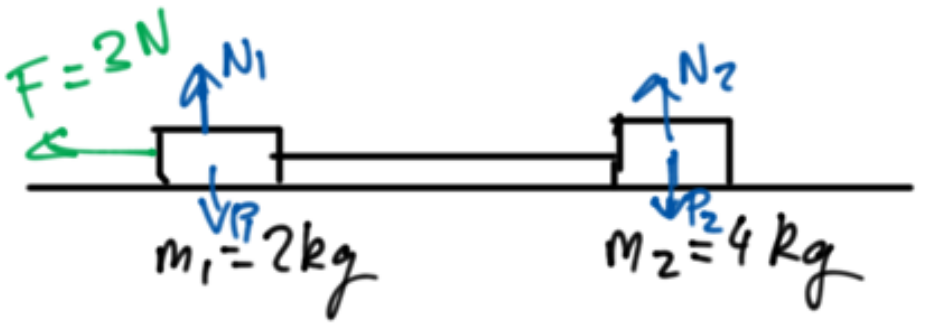
6) Si $mg = 2T \sin\alpha$

$$\alpha = \arcsin \frac{mg}{2T} = \arcsin \frac{2 \cdot 10}{2 \cdot 15}$$

$$\alpha = 41'81°$$

Si el ángulo es menor, la cuerda se rompe.

③

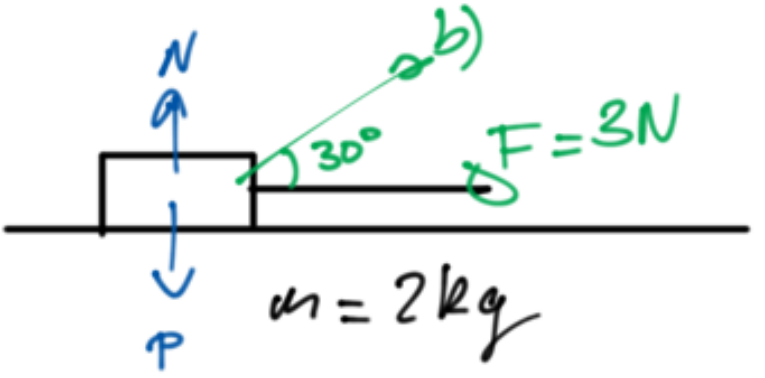


a) $F = (m_1 + m_2)\,a$; $a = \dfrac{F}{m_1 + m_2} = \dfrac{3}{2+4} = \dfrac{3}{6} = \dfrac{1}{2}\ m/s^2$

b) $a = 0.5\ m/s^2$, $m = 4\ kg$

$F = 4 \cdot 0.5 = 2N$

(4)



a) $F = ma$

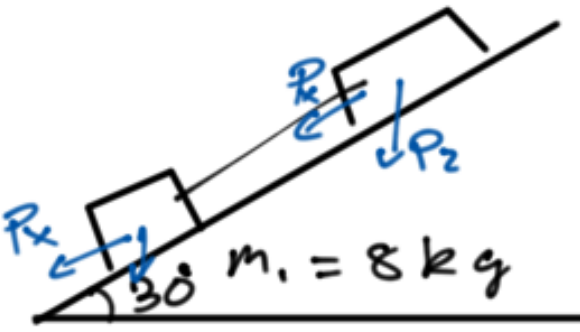
$$a = \frac{F}{m} = \frac{3}{2} = 1'5\ m/s^2$$

b) $F_x = F\cos 30° = m\,a$

$$a = \frac{F\cos 30}{m} = \frac{3 \cdot \sqrt{3}/2}{2} = \frac{3\sqrt{3}}{4}$$

$$= 1'3\ m/s^2$$

5



$m_2 = 4\,kg$

a) $P_{1x} = P_1 \cdot \sin\alpha = m_1 g \sin\alpha$

$P_{2x} = P_2 \cdot \sin\alpha = m_2 g \sin\alpha$

$\sum \vec{F} = m\vec{a}$; $P_{1x} - T + P_{2x} + T = m\,a$

$m_1 g \sin 30° + m_2 g \sin 30 = (m_1 + m_2)\,a$

$(m_1 + m_2)\, g \sin 30° = (m_1 + m_2)\, a$

$a = g \sin 30$

$a = 5\ m/s^2$

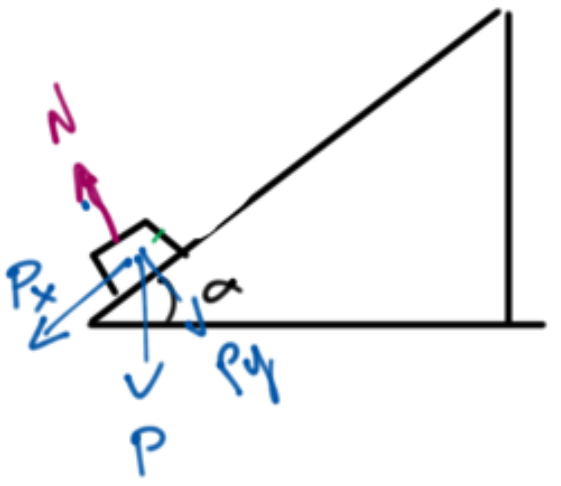
No depende de la masa.

6) $P_{1x} - T = m_1 a$

$T = m_1 a - m_1 g \cos\alpha$

$T = 8 \cdot 5 - 8 \cdot 10 \cdot \frac{1}{2} = 0\ N$

6



$a = 20°$

$v_0 = 2'5\ m/s$

$g = 10\ m/s^2$

a) La única fuerza que interviene en el ejex es P_x

$$P_x = m g \sin 20°$$

$$\sum \vec{F} = m\, a$$

$$m g \sin 20° = m \cdot a$$

$$a = g \sin 20 = 3'42\ m/s^2$$

b) $$\Delta x = v_0 t + \frac{1}{2} a t^2$$

$$\Delta v = a t \ ,$$

Como se frena, $\Delta v = -2'5\ m/s$

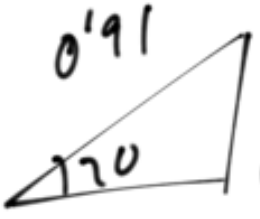
$$t = \frac{\Delta v}{a} = \frac{-2'5}{-3'42} = 0'73\ s$$

$$\Rightarrow \Delta x = 2'5\,t - \frac{1}{2}\,3'42\,t^2$$

c) Recorre hasta detenerse:

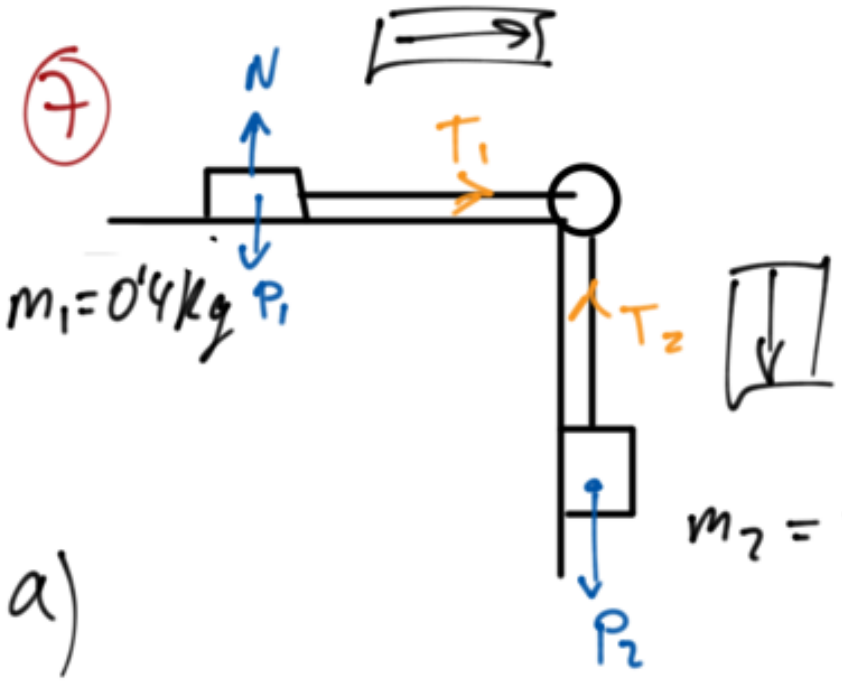
$$\Delta x = 2'5 \cdot 0'73 - \frac{3'42}{2}\,(0'73)^2$$

$$\Delta x = 0'91\ m$$



$$\rightsquigarrow h = 0'91 \cdot \sin 20 = 0'31\ m$$

7



$g = 10\ m/s^2$

$T_1 = -T_2 = T$

a)

② $P_2 + T_2 = m_2 a$; $m_2 g - T = m_2 a$

① $T_1 = m_1 a$ $\quad T = m_1 a$

$m_2 g = (m_1 + m_2) a$

$$a = \frac{m_2 g}{(m_1 + m_2)} = \frac{0'6 \cdot 10}{0'4 + 0'6} = 6\ m/s^2$$

$T = 0'4 \cdot 6 = 2'4\ N$

b) $F_x = ?$ Si $v_0 = 0$

$\Delta x = 1\,m$

$\Delta t = 0'82\,s$

$$\Delta x = v_0 t + \frac{1}{2} a t^2$$

$$a = \frac{2\Delta x}{t^2} = \frac{2 \cdot 1}{0'82^2} = 2'97\ m/s^2$$

$P_2 - F = m a$ ($= m_1 + m_2$)

$$F = P_2 - m a = m_2 g - (m_1 + m_2)\, 2'97 =$$

$$= 0'6 \cdot 10 - (0'4 + 0'6)\, 2'97$$

$$= 3'03\ N$$

$P_2 - T = m_2 a$

$T - F = m_1 a$

$$T = m_1 a + F = 0'4 \cdot 2'97 + 3'03$$

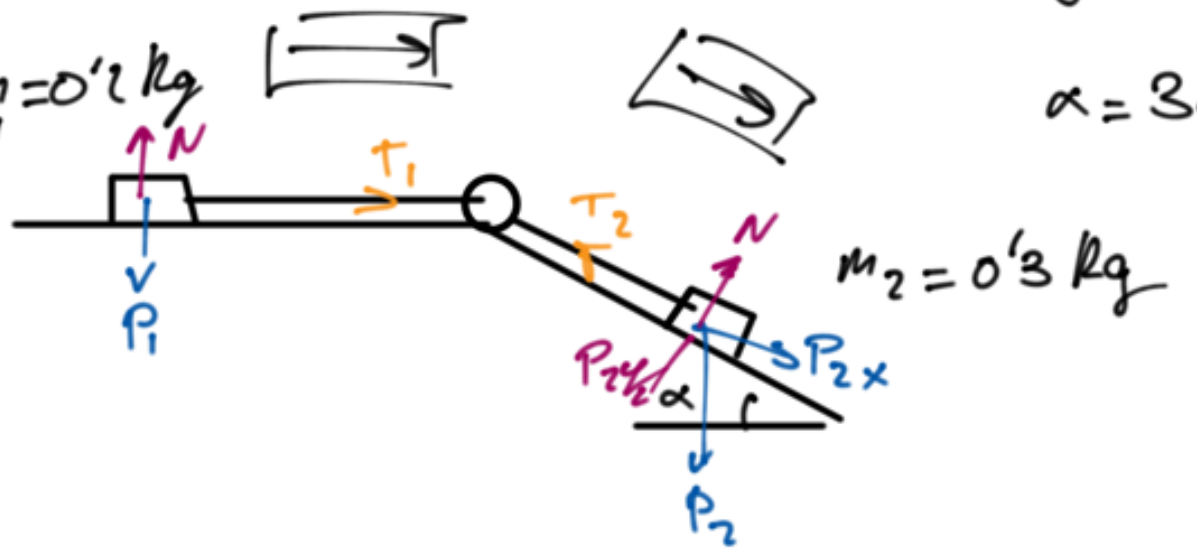
$$T = 4'22\ N$$

c)

⑧

$g = 10\ m/s^2$

$\alpha = 30°$



a)

$T_1 = -T_2 = T$

① $T_1 = m_1 a$; $T = m_1 a$

② $P_{2x} + T_2 = m_2 a$; $m_2 g \sin\alpha - T = m_2 a$

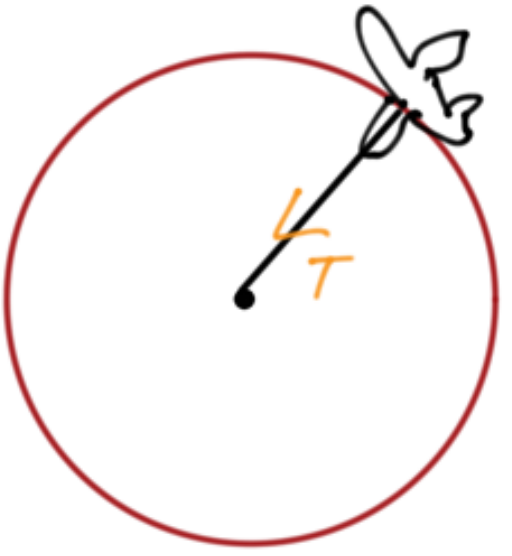
$m_2 g \sin\alpha = (m_1 + m_2) a$

$$a = \frac{m_2 g \sin\alpha}{m_1 + m_2} = \frac{0'3 \cdot 10 \cdot \sin 30}{0'3 + 0'2} =$$

$$= \frac{1'5}{0'5} = 3\ m/s^2$$

b) $T = 0'2 \cdot 3 = 0'6\ N$

9



$g = 10\ m/s^2$

$m = 0'5\ kg$

$R = 6\ m$

$T = 4$

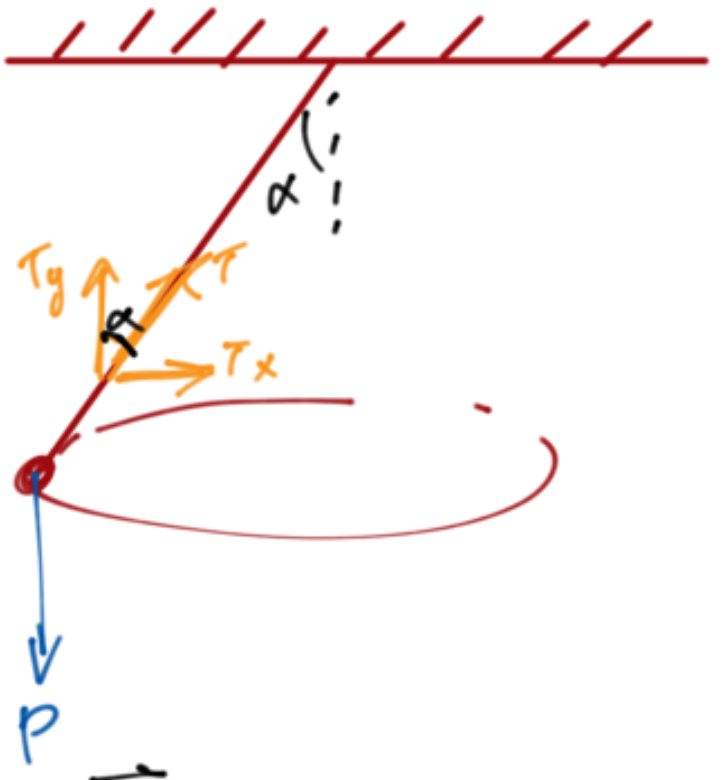
$$\omega = \frac{2\pi}{T}$$

$$v = R\omega = R \cdot \frac{2\pi}{T}$$

$$T = F_c = m\frac{v^2}{R} = \frac{m R^2 \omega^2}{R} = \frac{m R 4\pi^2}{T^2} =$$

$$= \frac{0'5 \cdot 6 \cdot 4\pi^2}{4^2} = 7'4\ N$$

10



$g = 10\ m/s^2$
$m = 0'225\ kg$
$R = 0'5\ m$
$\alpha = 30^\circ$

$$\vec{T} = T_x\,\vec{i} + T_x\,\vec{j} = T\sin\alpha\,\vec{i} + T\cos\alpha\,\vec{j}\ (N)$$

$$\vec{P} = -mg\,\vec{j}$$

$$\sum \vec{F} = m\vec{a}$$

Eje y: $T\cos\alpha - mg = 0$

Eje x: $T\sin\alpha = F_c\ ;\ T\sin\alpha = m\frac{v^2}{R}$

$$T = \frac{mg}{\cos\alpha} = \frac{0'225 \cdot 10}{\cos 30} = 2'60\ N$$

$$T \sin\alpha = m \frac{v^2}{R}$$

$$v = \sqrt{\frac{T \sin\alpha \, R}{m}} = \sqrt{\frac{2'60 \cdot \sin 30 \cdot 0'5}{0'225}} =$$

$$= 1'70 \text{ m/s}$$

b) $T = 2'60 \text{ N}$

c) La resultante es centrípeta

$$F = m \frac{v^2}{R} = \frac{0'225 \cdot 1'7^2}{0'5} = 1'3 \text{ N}$$